FPGA Tactical Controller for StarCraft II Micro Scenarios

Target Tier: Tier 'A' (100%) **Course:** EECS 3216 (Fall 2026) **Topic:** Datapath / Control / VGA

Group Members: [Student 1 Name] ([ID]), [Student 2 Name] ([ID]), [Student 3 Name] ([ID]), [Student 4 Name] ([ID])

1. Overview and Difficulty Justification

We propose a SystemVerilog controller demonstrated in fixed-army StarCraft II (SC2) scenarios without production. A PC adapter exchanges observations/commands over UART; the FPGA assesses strength, selects advance/attack/retreat/hold behavior, and allocates targets. SC2 renders gameplay on the PC; FPGA VGA shows a simple top-down diagram of colored markers and text. Tier 'A' combines original UART, memory coordination, pipelined arithmetic, dependent reservations, interacting FSMs, and video output.

2. Target Hardware and Peripherals (DE10-Lite Only)

- **Onboard Resources (MAX 10 `10M50DAF484C7G`):** 50 MHz clock (`PIN_P11`); M9K RAM; multipliers; 25.2 MHz VGA PLL; GPIO UART RX/TX; 4-bit-per-channel VGA; `KEY0` reset, `KEY1` step; `SW4` auto/step, `SW3:0` unit selection; `LEDR9:0` status; `HEX5:0` results/cycles.
- **External Hardware:** SC2 host PC, 3.3 V USB-to-UART adapter/common-ground wiring, and VGA monitor/cable (640×480 at 60 Hz). USB-Blaster programs the FPGA.

3. Combinational and Sequential Logic Architecture

- **Combinational Logic:** Coordinate subtractors, distance/strength multipliers, range/waypoint comparators, reservation arithmetic, minimum selector, checksum adder, and display decoders. Unsigned 16-bit positions/reach have six fractional bits; 17-bit signed differences feed a 35-bit distance-squared sum. Health/damage use 16 bits/four fractional bits; reservations 20 bits, strength sums 36 bits; comparisons are widened.
- **Sequential Logic and FSMs:** UART synchronizer/RX/TX FSMs; M9K memories; four distance/ranking stages; accumulators; scan/cycle counters; retreat latch; master FSM (`RECEIVE, VALIDATE, ASSESS, ALLOCATE, REPLY, ERROR`). FSM patterns [1,3] inform `ADVANCE, ATTACK, RETREAT, HOLD`. Scans drain before reservation writes; computation locks RAM. VGA counters and a frame-boundary handshake synchronize stable display data.

4. Concrete Specifications and Testable Features

- **Protocol:** 115,200-baud UART, 8-N-1, 8-bit additive checksums. Snapshots carry IDs, 0–16 units/side, indices, health/max-health, positions, damage, readiness, and objective/retreat waypoints. Return state, commands, targets/waypoints, and cycles. Reject malformed packets/counts, checksum errors, and 100 ms inter-byte timeouts; round reset clears history.
- **Tactical Decisions:** Use $S = \sum_i H_i D_i$ over living units (health H_i, damage/attack D_i). Retreat at enemy strength $\geq 2S_{\text{friendly}}$ or health $\leq 25\%$ of initial squad maximum; latch retreat until reset, holding at safety. Otherwise attack when ready/in range, wait during cooldown in range, or advance to the objective. Empty squads hold; arrival tolerance is one map unit. Score/cutoffs are experimental parameters.
- **Target Allocation:** Adapt weakest-target/no-overkill ideas [2]: process ready attackers in index order; rank living in-range targets by positive health after reservations, squared distance, then index. Reserve damage before the next attacker; unassigned attackers hold. Reset reservations per snapshot. Support Marines versus Zerglings on open terrain without upgrades/abilities, with per-attacker damage/reach constant across targets.
- **SC2 Demo and Benchmarks:** Select/adapt AI Arena micro-map rounds [6] to 4–8 units/side; pin map/client versions and verify spawns, resets, counts, and terrain. Single-step SC2 waits for hardware; the host maps tags and validates commands. Demonstrate approach, allocation, and retreat. Vary starts/health/force balance [4]; compare Attack-Closest [2] and pinned MarineMicroBot [5] with matched scenarios/opponent/step intervals. Record outcomes, survivors, health, game loops, and compute/round-trip latency. VGA shows markers, target/waypoint highlights, and state text; HEX/LEDs show results/cycles/status.
- **Originality:** HDL, adapter, and tests will be original; algorithms credit [1–3], SDK/example code [5], maps [6]. Only PLL IP is planned; RAM/multipliers are inferred. Pin/timing source: Terasic's DE10-Lite User Manual.

5. Verification Plan and Team Task Allocation

- **Student 1 (UART/Host):** ModelSim protocol/error tests, loopback, SC2 tag/command mapping and round resets.
- **Student 2 (Datapath):** Check fixed-point limits, thresholds, and ties against an independent reference.
- **Student 3 (FSMs/Allocation):** Check transitions, retreat persistence, reservations, empty/cooldown cases, and bit-exact results on 1,000 randomized snapshot sequences.
- **Student 4 (VGA/Integration):** Verify video timing and clock crossings; test maps and benchmarks; report FPGA resources, timing, and gameplay outcomes.

References [1] I. Millington, AI for Games, 3rd ed., 2019, Sec. 5.3, pp. 314–317.
[2] D. Churchill and M. Buro, Portfolio Greedy Search and Simulation for Large-Scale Combat in StarCraft, 2013, Secs. II–III.
[3] E. A. Lee and S. A. Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, 2nd ed., 2017, Sec. 3.3.
[4] B. Ellis et al., SMACv2: An Improved Benchmark for Cooperative Multi-Agent Reinforcement Learning, 2023.
[5] Blizzard, s2client-api: MarineMicroBot (examples/common/bot_examples.cc); SC2 protocol: single-step mode.
[6] AI Arena, micro-map downloads and micro-ladder-maps source; versions selected after compatibility testing.